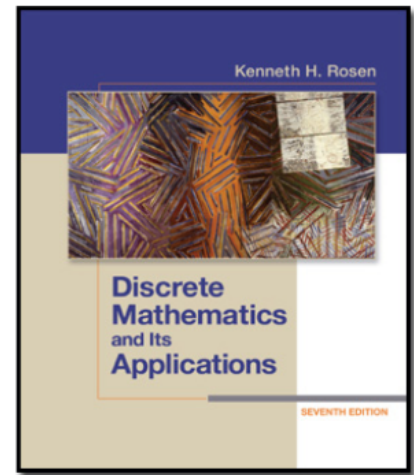




Chapter 1, Part I

Propositional Equivalences

Chapter Summary

- Propositional Logic
- Predicate Logic
- Proofs

Section 1.3 of Rosen's Book

Section Summary

- Tautologies, Contradictions, and Contingencies.
- Logical Equivalence
 - Important Logical Equivalences
 - Showing Logical Equivalence
- Normal Forms (*optional, covered in exercises in text*)
 - Disjunctive Normal Form
 - Conjunctive Normal Form
- Propositional Satisfiability
 - Sudoku Example

Tautologies, Contradictions, and Contingencies

A *tautology* is a proposition which is always true.

- Example: $p \vee \neg p$

Tautologies, Contradictions, and Contingencies

A *contradiction* is a proposition which is always false.

- Example: $p \wedge \neg p$

Tautologies, Contradictions, and Contingencies

A *contingency* is a proposition which is neither a tautology nor a contradiction, such as p

Tautologies, Contradictions, and Contingencies

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Logically Equivalent

p and q are *logically equivalent* if $p \leftrightarrow q$ is a tautology.

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p and q are *logically equivalent* if $p \leftrightarrow q$ is a tautology.

$\neg p \vee q$ is equivalent to $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

$$(\neg p \vee q) \equiv (p \rightarrow q)$$

De Morgan's Laws



Augustus De Morgan

1806-1871

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

This truth table shows that De Morgan's Second Law holds.

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Key Logical Equivalences

- Identity Laws: $p \wedge T \equiv p$ $p \vee F \equiv p$
- Domination Laws: $p \vee T \equiv T$, $p \wedge F \equiv F$
- Idempotent laws: $p \vee p \equiv p$, $p \wedge p \equiv p$
- Double Negation Law: $\neg(\neg p) \equiv p$
- Negation Laws: $p \vee \neg p \equiv T$, $p \wedge \neg p \equiv F$

Key Logical Equivalences (*cont*)

- Commutative Laws: $p \vee q \equiv q \vee p$, $p \wedge q \equiv q \wedge p$
- Associative Laws: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
 $(p \vee q) \vee r \equiv p \vee (q \vee r)$
- Distributive Laws: $(p \vee (q \wedge r)) \equiv (p \vee q) \wedge (p \vee r)$
 $(p \wedge (q \vee r)) \equiv (p \wedge q) \vee (p \wedge r)$
- Absorption Laws: $p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$

More Logical Equivalences

TABLE 7 Logical Equivalences Involving Conditional Statements.

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

TABLE 8 Logical Equivalences Involving Biconditional Statements.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Equivalence Proofs

Example: Show that $\neg(p \vee (\neg p \wedge q))$
is logically equivalent to $\neg p \wedge \neg q$

Solution:

$\neg(p \vee (\neg p \wedge q))$	$\equiv$	$\neg p \wedge \neg(\neg p \wedge q)$	by the second De Morgan law
	$\equiv$	$\neg p \wedge [\neg(\neg p) \vee \neg q]$	by the first De Morgan law
	$\equiv$	$\neg p \wedge (p \vee \neg q)$	by the double negation law
	$\equiv$	$(\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the second distributive law
	$\equiv$	$F \vee (\neg p \wedge \neg q)$	because $\neg p \wedge p \equiv F$
	$\equiv$	$(\neg p \wedge \neg q) \vee F$	by the commutative law for disjunction
	$\equiv$	$(\neg p \wedge \neg q)$	by the identity law for F

Equivalence Proofs

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$
is a tautology.

Solution:

$$\begin{aligned}(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \text{by truth table for } \rightarrow \\ &\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{by the first De Morgan law} \\ &\equiv (\neg p \vee p) \vee (\neg q \vee q) && \text{by associative and} \\ &&& \text{commutative laws} \\ &&& \text{laws for disjunction} \\ &\equiv T \vee T && \text{by truth tables} \\ &\equiv T && \text{by the domination law}\end{aligned}$$

Disjunctive Normal Form (*optional*)

- A propositional formula is in *disjunctive normal form* if it consists of a disjunction of $(1, \dots, n)$ disjuncts where each disjunct consists of a conjunction of $(1, \dots, m)$ atomic formulas or the negation of an atomic formula.
 - Yes
$$A$$
$$(A \wedge \neg B) \vee (\neg C \wedge D \wedge E)$$
$$(C \wedge D) \vee E$$
 - No
$$C \wedge (D \vee E)$$
- Disjunctive Normal Form is important for the circuit design methods.

Disjunctive Normal Form (optional)

Example: Find the Disjunctive Normal Form (DNF) of

$$(p \vee q) \rightarrow \neg r$$

Solution: This proposition is true when r is false or when both p and q are false.

$$(\neg p \wedge \neg q) \vee \neg r$$

Conjunctive Normal Form (optional)

- A compound proposition is in *Conjunctive Normal Form* (CNF) if it is a conjunction of disjunctions.
- Every proposition can be put in an equivalent CNF.
- Conjunctive Normal Form (CNF) can be obtained by eliminating implications, moving negation inwards and using the distributive and associative laws.
- Important in resolution theorem proving used in artificial Intelligence (AI).
- A compound proposition can be put in conjunctive normal form through repeated application of the logical equivalences covered earlier.

Conjunctive Normal Form (optional)

Example: Put the following into CNF:

$$\neg(p \rightarrow q) \vee (r \rightarrow p)$$

Solution:

1. Eliminate implication signs:

$$\neg(\neg p \vee q) \vee (\neg r \vee p)$$

2. Move negation inwards; eliminate double negation:

$$(p \wedge \neg q) \vee (\neg r \vee p)$$

3. Convert to CNF using associative / distributive laws

$$(p \vee \neg r \vee p) \wedge (\neg q \vee \neg r \vee p)$$

Propositional Satisfiability

- A compound proposition is *satisfiable* if there is an assignment of truth values to its variables that make it true.
- When *no such assignments exist*, the compound proposition is *unsatisfiable*.

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- When no such assignments exist, the compound proposition is *unsatisfiable*.

A compound proposition is *unsatisfiable* if and only if *its negation is a tautology*.

Questions on Propositional Satisfiability

Example: Determine the satisfiability of the following compound propositions:

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$$

Questions on Propositional Satisfiability

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Solution:

Satisfiable. Assign **T** to p, q, and r.

Questions on Propositional Satisfiability

Example: Determine the satisfiability of the following compound propositions:

$$(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Questions on Propositional Satisfiability

Example: Determine the satisfiability of the following compound propositions:

$$(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Solution:

Satisfiable. Assign **T** to p and **F** to q.

Questions on Propositional Satisfiability

Example: Determine the satisfiability of the following compound propositions:

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Solution:

Not satisfiable.

Check each possible assignment of truth values to the propositional variables and **none** will make the proposition true.

Notation

$\bigvee_{j=1}^n p_j$ is used for $p_1 \vee p_2 \vee \dots \vee p_n$

$\bigwedge_{j=1}^n p_j$ is used for $p_1 \wedge p_2 \wedge \dots \wedge p_n$

Needed for the next example.

Sudoku

- A **Sudoku puzzle** is represented by a 9×9 grid made up of nine 3×3 subgrids, known as **blocks**. Some of the 81 cells of the puzzle are assigned one of the numbers 1,2, ..., 9.
- The puzzle is solved by assigning numbers to each blank cell so that every row, column and block contains each of the nine possible numbers.

- Example

	2	9				4		
			5			1		
	4							
				4	2			
6							7	
5								
7			3					5
	1			9				
							6	

Encoding as a Satisfiability Problem

- Let $p(i,j,n)$ denote the proposition that is true when the number n is in the cell in the i th row and the j th column.
- There are $9 \times 9 \times 9 = 729$ such propositions.
- In the sample puzzle $p(5,1,6)$ is true, but $p(5,j,6)$ is false for $j = 2,3,\dots,9$

Encoding (cont)

- For each cell with a given value, assert $p(i,j,n)$, when the cell in row i and column j has the given value.
- Assert that every row contains every number.

$$\bigwedge_{i=1}^9 \bigwedge_{n=1}^9 \bigvee_{j=1}^9 p(i, j, n)$$

$[(p(1,1,1) \vee p(1,2,1) \vee \dots p(1,9,1)) \text{and} (p(1,1,2) \vee \dots p(1,9,2)) \text{and} \dots]$ and
 $[(p(2,1,1) \vee p(2,2,1) \vee \dots p(2,9,1)) \text{and} (p(2,1,2) \vee \dots p(2,9,2)) \text{and} \dots]$ and
...
 $[(p(9,1,1) \vee p(9,2,1) \vee \dots p(9,9,1)) \text{and} (p(9,1,2) \vee \dots p(9,9,2)) \text{and} \dots]$

Encoding (cont)

- For each cell with a given value, assert $p(i,j,n)$, when the cell in row i and column j has the given value.
- Assert that every column contains every number.

$$\bigwedge_{j=1}^9 \bigwedge_{n=1}^9 \bigvee_{i=1}^9 p(i, j, n)$$

$[(p(1,1,1) \vee p(2,1,1) \vee \dots \vee p(9,1,1)) \wedge (p(1,1,2) \vee \dots \vee p(9,1,2)) \wedge \dots]$ and
 $[(p(1,2,1) \vee p(2,2,1) \vee \dots \vee p(9,2,1)) \wedge (p(1,2,2) \vee \dots \vee p(9,2,2)) \wedge \dots]$ and
...
 $[(p(1,9,1) \vee p(2,9,1) \vee \dots \vee p(9,9,1)) \wedge (p(1,9,2) \vee \dots \vee p(9,9,2)) \wedge \dots]$

Encoding (cont)

- Assert that each of the 3×3 blocks contain every number.

$$\bigwedge_{r=0}^2 \bigwedge_{s=0}^2 \bigwedge_{n=1}^9 \bigwedge_{i=1}^3 \bigvee_{j=1}^3 p(3r + i, 3s + j, n)$$

(this is tricky - ideas from chapter 4 help)

$[(p(1,1,1) \vee p(1,2,1) \vee p(1,3,1)) \text{ and } (p(2,1,1) \vee p(2,2,1) \vee p(2,3,1)) \text{ and } (p(3,1,1) \vee p(3,2,1) \vee p(3,3,1)) \text{ and}$
 $(p(1,1,2) \vee p(1,2,2) \vee p(1,3,2)) \text{ and } (p(2,1,2) \vee p(2,2,2) \vee p(2,3,2)) \text{ and } (p(3,1,2) \vee p(3,2,2) \vee p(3,3,2)) \text{ and}$
 $\dots$
 $(p(1,1,9) \vee p(1,2,9) \vee p(1,3,9)) \text{ and } (p(2,1,9) \vee p(2,2,9) \vee p(2,3,9)) \text{ and } (p(3,1,9) \vee p(3,2,9) \vee p(3,3,9))]$ and
 $[(p(1,4,1) \vee p(1,5,1) \vee p(1,6,1)) \text{ and } (p(2,4,1) \vee p(2,5,1) \vee p(2,6,1)) \text{ and } (p(3,4,1) \vee p(3,5,1) \vee p(3,6,1)) \text{ and}$
 $(p(1,4,2) \vee p(1,5,2) \vee p(1,6,2)) \text{ and } (p(2,4,2) \vee p(2,5,2) \vee p(2,6,2)) \text{ and } (p(3,4,2) \vee p(3,5,2) \vee p(3,6,2)) \text{ and}$
 $\dots$
 $(p(1,4,9) \vee p(1,5,9) \vee p(1,6,9)) \text{ and } (p(2,4,9) \vee p(2,5,9) \vee p(2,6,9)) \text{ and } (p(3,4,9) \vee p(3,5,9) \vee p(3,6,9))]$ and
 $\dots$

Encoding (cont)

- Assert that no cell contains more than one number.
- Take the conjunction over all values of n, n', i , and j , where each variable ranges from 1 to 9 and $n \neq n'$, of $p(i, j, n) \rightarrow \neg p(i, j, n')$

Solving Satisfiability Problems

- To solve a Sudoku puzzle, we need to find an assignment of truth values to the 729 variables of the form $p(i,j,n)$ that makes the conjunction of the assertions true.
 - > *Those variables that are assigned T yield a solution to the puzzle.*
- A truth table can always be used to determine the satisfiability of a compound proposition. But this is too complex even for modern computers for large problems.
- There has been much work on developing efficient methods for solving satisfiability problems as many practical problems can be translated into satisfiability problems.